Hariprashad Ravikumar

PhD Candidate specializing in Physics-Based Modeling, Predictive Analytics, and AI for Science

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Summary

PhD physicist with 5+ years in HPC, ML for science, and physics-based simulation. Shipped production tools (Dash, CI/CD, Kubernetes, PyPI) at Western Digital. Seeking full-time roles in data science, ML engineering, or computational physics.

Experience

1. **HAMR Modeling and Simulation Intern**, Western Digital, San Jose, CA *(May 2026 – Present)*
   HAMR DCSNR Simulation Web Application (NIMBLE)
   - Built a full-stack interactive simulation app using Python, Dash, and Plotly with real-time parameter sweeps, multi-run comparison, and automated PowerPoint/Excel report generation; adopted by cross-functional media and sputtering engineering teams
   - Implemented auto-fit THMap curve fitting (skew-normal thermal profile + RTS noise) and physics model pipeline with grain noise and reader Gaussian PSF convolution to compute unsqueezed/squeezed DCSNR metrics
   - Leveraged IndexedDB API for client-side state persistence and one-time data handoff across browser tabs, enabling seamless cross portal integration other apps
   - Deployed via CI/CD pipeline in Jenkins with Kubernetes; packaged as a modular PyPI-compatible Python library integrated into WD's internal multi-tool engineering platform

   HAMR Physics Modeling
   - Developed a closed-form analytical model for HAMR switching probability and DCSNR from first principles, combining Néel-Arrhenius thermal activation and Stoner-Wohlfarth coherent rotation for $L1_0$ FePt grains; peer-reviewed publication manuscript in preparation
   - Validated model against Monte Carlo simulation and spin-stand data across grain ensembles with distributions of H_k, volume, and T_c
   - Quantified multi-write noise power accumulation and DCSNR degradation to provide areal density capability (ADC) and write endurance metrics, helping media design teams reduce adjacent track erasure (ATI & xTI)

2. **Graduate Research Assistant**, NM State University, Las Cruces, NM *(Aug 2021 – Present)*
   PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches
   - Generated 30,000+ high-fidelity synthetic data points by solving Partial Differential Equations with large-scale Hybrid Monte Carlo simulations (Markov Chain stochastic process) and built an end-to-end AI for Science pipeline to model the underlying physics, achieving over 98% predictive accuracy with symbolic regression machine learning.
   - Engineered parallelized Bayesian inference pipelines across multi-CPU architectures to fit non-linear, high-dimensional models; utilized automatic differentiation and full covariance matrices to ensure exact error propagation and numerical stability in multi-stage workflows (jackknife resampling)

Independent Collaborations

1. **Los Alamos National Laboratory** - Lattice Quantum Chromodynamics *(May 2024 – Present)*
   - Accelerated multi-terabyte scientific calculations by developing and optimizing parallelized C++ CUDA kernels for GPU-accelerated HPC clusters (NERSC Perlmutter), significantly reducing runtime for large-scale Hamiltonian Monte Carlo simulations.
   - Managed and executed 75,000+ CPU/GPU compute hours by designing and deploying custom SLURM workflows for large-scale job orchestration
2. **North Carolina State University** - Mathematical Physics *(Dec 2020 - Present)*
   - Implemented and managed Mathematica symbolic computation workflows on HPC clusters to analyze complex algebraic structures and symmetry constraints.

Selected Publications

Google Scholar

1. C.-R. Ji and **H. Ravikumar**, *"Interpolating conformal algebra in (1+1) dimensions,"* **Physical Review D** 113, 096018 (May 2026). DOI: 10.1103/prlq-4j1l.

Technical Projects

1. **AI-DataScience-Lab: Full-Stack Data Product** — GitHub | Live App
   - Prototyped and deployed a scalable full-stack data tool (Python, Flask, Azure) to deliver on-demand data-driven insights and actionable recommendations, integrating an OpenAI API to demonstrate automated data storytelling and data interpretation.
   - Implemented an end-to-end statistical analysis pipeline using Python (Pandas) for data cleaning, statistical modeling (Scikit-learn) for regression, and Matplotlib for data visualization.
2. **Physics Based Monte Carlo Simulation** — GitHub
   - Developed a Physics-Based Simulation from scratch to generate synthetic lattice gauge configurations using Monte Carlo methods on HPC clusters, validating the generated data against known analytical benchmarks.

Technical Skills

Programming	Python, C++, CUDA, Bash, SQL, Lua, HTML/CSS, YAML
ML & APIs	PyTorch, TensorFlow, Scikit-learn, Pandas, Dash, Plotly, Flask, FastAPI, Numba
Cloud & MLOps	Azure, AWS (S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD, Git, SLURM
Methods & HPC	Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)

Education

PhD in Physics, New Mexico State University, USA — *Aug 2021 – Dec 2026 (expected)*
MS in Physics, New Mexico State University, USA — *Aug 2021 – May 2024*
MSc in Physics, National Institute of Technology Jalandhar, India — *July 2019 – May 2021*
BSc in Physics, Dr. N.G.P. Arts and Science College, India — *June 2015 – May 2018*

Certifications

- Getting Started with Accelerated Computing in CUDA C/C++ by NVIDIA
- Fundamentals of Accelerated Computing with CUDA Python by NVIDIA
- Advanced Learning Algorithms by DeepLearning.AI
- Supervised Machine Learning: Regression and Classification by DeepLearning.AI
- Google Advanced Data Analytics Professional Certificate

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium to conduct my independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Leadership

- **Vice President**, Physics Graduate Student Organization (NMSU) — *Aug 2025 – Present*
  Organized professional development events and served as the primary liaison between 40+ graduate students and faculty.

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics, Electromagnetic Theory I & II